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Assessing the climate change and its impact on rice yields of Haridwar district using PRECIS RCM data

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Abstract In the past few decades, continuous intervention with the environmental landscape in the form of land use practices (water diversions, deforestation, local agriculture practices, industrialization etc.) in the Haridwar district of Uttarakhand, India, has impacted the region on various accounts. This is likely to be further aggravated in view of the increasing variability in the weather condition. Thus, there is urgent need felt to quantitatively asses the future climatic scenario for initiating and effectively undertaking the adaptation strategies for safe and sustained agricultural growth. For undertaking this study, PRECIS RCM data was downloaded for Roorkee grid station for the period (1961–2090). This PRECIS RCM data was biased corrected by merging two different bias correction methods which involves correcting mean and standard deviation simultaneously. Observed and bias corrected daily data was tested for their significance at 95% probability of occurrence employing various statistical methods such as correlation coefficient, mean bias error, normalized mean squared error, and Z and F statistical tests. These tests revealed that the difference between the observed and bias corrected PRECIS RCM data was insignificant. Trend analysis was done using Mann-Kendall's test and Theil Sen's Slope. Analysis of the bias corrected data revealed that the Roorkee station will, in general, record increasing trends in rainfall (at 5.83 mm/year), maximum temperature (at 0.05 °C/year), and minimum temperature (at 0.04 °C/year). This indicates that by 2090 AD present normal rainfall of 1006 mm/year will rise to 1447 mm/year; maximum temperature of 30.1 °C will rise to 33.9 °C and minimum temperature of 17.5 °C will rise to 20.5 °C by 2090. This weather scenario cautions at the future agronomic practices for sustained productivity. Grain yield of rice cv. *Sharbati* for the period 2014–2090 was simulated by running DSSAT CERES rice model with PRECIS RCM weather data of Haridwar district and RCP CO_2 emission scenarios. The simulation result showed that if the

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crop is grown with the existing soil and crop management conditions, rice productivity of Haridwar in general will decrease 31.7 kgs/ha/year. This could be attributed to the climate change and decreased rice yields from increased maximum temperature. In the case of RCP 2.6 and RCP 4.5, the effect of CO_2 on grain yields is more pronounced compared to RCP 6.0 and RCP 8.5. With the higher CO_2 concentrations (above 450 ppm), the rice crop cv. Sharbati did not show much CO_2 fertilization effect. DSSAT CERES rice simulations under climate change and higher CO_2 concentrations showed that the yields will decline with the advancing climate change but CO_2 intervention will compensate the loss in yield.

1 Introduction

Increased concentration of anthropogenic greenhouse gases (GHG) has led to climate change and has emerged in the world as a key concern environmentally and economically. The alarmingly increasing GHG concentrations results in increased temperature and thereby influencing the cyclic pattern of weather condition. In addition to the normal changes, the frequency of occurrence of extreme weather events has also been influenced due to greenhouse gases (Ahsan et al. 2010; Yu et al. 2010). The Intergovernmental Panel on Climate Change (IPCC) in its fourth assessment report (AR4) states that the world over-industrialization has caused the planet to warm by about 1 °C in the twenty-first century (Solomon et al. 2007). Doubling of carbon dioxide emission in the atmosphere was caused due to biotic and abiotic reasons; this trend is projected to be further intensified and may cause increase in global warming by about 3 °C (Harley et al. 2006). This climatic changes directly or indirectly affect the agricultural sector and the productivity is also likely to be adversely affected (Gornall et al. 2010; Roudier et al. 2011). The agrarian countries like India are worse hit with such changes where 52% of the livelihood comes from agriculture sector and its allied activities. Agriculture GDP's contribution to Indian economy is 18.1% as per 2016 statistics (Srivastava et al. 2016). Based on over hundred years of data collected by the India Meteorological Department (IMD) over the Indian region have shown warming of about 0.5 °C per 100 years in an all-India average basis (Subash et al. 2013). Future projections of climate change using regional climate model, run by the Indian Institute of Tropical Meteorology (IITM) with different IPCC emission scenarios, indicate the temperature increase of about 3–5 °C by turn of the century (Kumar et al. 2006).

In the past few decades, changes in the environmental landscape of Haridwar District (Uttarakhand, India) in the form of changes in the land use practices (water resource diversions, deforestation, intensified agriculture practices, industrialization etc.) have caused large-scale impacts on the climate variability in the region. Tripathi et al. 2007 using the weather data for the period 1979–2008 reported a rise in average temperature at 0.025 °C/year for Roorkee (Haridwar District) station (Tripathi et al. 2007). This change process is expected to be further accelerated in the future. Therefore, there is a need to quantify the climate change process of a given location and simulate the amplitude and develop local adaptation strategies for sustained growth of agriculture.

Data of the PRECIS Regional Climate Model developed by the UK's Hadley Centre for Climate Prediction and Research, to generate detailed climate change predictions at 50 ×

50 km ($0.44^0 \times 0.44^0$) and 25 × 25 km ($0.22^0 \times 0.22^0$) grid obtained from IITM (Indian Institute of Tropical Meteorology), is a very good tool for long-term weather projection of a given location. The high resolution in PRECIS RCM as compared to General Circulation Models (GCM) gives detailed information on the vulnerability to climate change of a site (Grotch and MacCracken 1991; Fowler et al. 2007; Maraun et al. 2010; Field et al. 2014). Therefore, PRECIS RCM data was used to predict rainfall (mm), maximum temperature (°C), and minimum temperature (°C) for Haridwar district of Uttarakhand India for the period 1979–2090. Although the PRECIS RCM model-simulated results present the future climate scenario, but it is substantially different from the actual. The difference could be minimized using bias correction methods such as linear correction approach (Lenderink et al. 2007), non-linear correction approach (Leander and Buishand 2007), γ distribution correction approach (Hay et al. 2002), and empirical distribution correction approach (Wood et al. 2002).

Climate change impact on crops can be studied using crop simulation models which are found to be effective means to translate the field conditions onto your desktop with various options to modify and also analyze the effect of various climatic factors on crop growth and yield. DSSAT models (Decision Support System for Agro technology Transfer) have been widely used for yield gap analysis, decision making and planning, strategic and tactical management decisions, climate change impact studies etc (Jones et al. 2003). In India, several studies have used DSSAT for impact assessment of climatic change (Rao et al. 2016; Srivastava et al. 2016; Singh et al. 2017). DSSAT model was evaluated under various scenarios for rice crop and observed that the actual and simulated yields did not show much deviation (Satapathy et al. 2014).

Everywhere, there is an emphasis on climate change and CO_2 also has a major role to play in such a scenario and therefore it is necessary to evaluate CO_2 elevation for the coming years. With this objective, the future weather for Haridwar district was forecasted using PRECIS RCM, given as weather input to CERES Rice model under four RCP emission scenarios to study their impact on rice grain yield for the period 2014–2090.

2 Methodology

PRECIS RCM data (rainfall, maximum temperature, and minimum temperature) was obtained from the Indian Institute of Tropical Meteorology (IITM), Pune in binary format. This data is converted into *netcdf* format using FORTRAN program and thereafter the daily data of rainfall, maximum temperature, and minimum temperature was extracted for Roorkee location (29° 52′ N and 77° 54′ S) for the period 01/01/1979–31/12/2090 using Arc GIS 9.3. PRECIS RCM daily weather data is available for 360 days/year considering 30 days per month only. Therefore, the daily data was adjusted to match the Gregorian calendar days/month taking average ofthe last five days of the months and accounting the same for the 31st day. No adjustment was made for the months having only 30 days/month. February is either 28 or 29 days/month, therefore one or two days were deleted from the data of 30 days. Data for the period 1979–2008 was used for calibration; data for the period 2009–2013 was used for validation; whereas, the data for the period 2014–2090 was bias corrected and used for climate and crop yield forecasting.

2.1 Steps adopted for bias correction of PRECIS RCM data are described below:

2.1.1 Correction for standard deviation (Leander and Buishand 2007)

$$P_{SD_{corr}} = \overline{P}_{obs} + \left(\frac{\sigma(P_{obs})}{\sigma(P_{rcm})}\right) \times \left(P_i - \overline{P_{obs}}\right) + \left(\overline{P_{obs}} - \overline{P_{rcm}}\right) \quad (1)$$

where
$P_{SD_{corr}}$=Bias corrected precipitation or temperature (daily/monthly)
$\overline{P}_{obs}$=Mean daily observed precipitation or temperature
$\overline{P_{rcm}}$=Mean daily PRECIS precipitation or temperature
$\sigma(P_{obs})$=Standard deviation of the observed precipitation or temperature
$\sigma(P_{rcm})$= Standard deviation of the PRECIS precipitation or temperature
P_i=Daily or monthly precipitation or temperature of PRECIS data

2.1.2 Correction for mean (Hay et al. 2002)

$$P_{mean\ \ corr} = P_{SD_{corr}} \times \left(\frac{\overline{P_{obs}}}{\overline{P_{rcm}}}\right) \quad (2)$$

2.1.3 Correction for other biases

In the first step, there is chance of generating negative values which is not desirable especially in the case of precipitation and even in temperatures where it is either abnormally high or low. In order to check such biases, the following conditions were employed. The precipitation data of a given day was programmed in such a way that the negative values were automatically replaced with zero.

2.2 Statistical evaluation

Calibrated and validated data was evaluated by adopting normalized root mean square error (NMSE, mean bias error (MBE) *Z* test, forecast accuracy (FA), ratio score, or hit score). Using all the above-mentioned evaluation measures, the forecast is verified and further climate change assessment was done using Mann-Kendall's trend test (MKTT).

2.3 Trend analysis

To assess future climate change, it is necessary to observe the variability and magnitude of change. To assess the trend and its magnitude, MKTT (Mann 1945; Kendall 1946) is used, which is one of the widely used non-parametric tests to detect significant trends in time series, along with Theil Sen's slope estimator (Sen 1968; Theil 1992). MKTT, being a function of the ranks of the observations rather than their actual values, is not affected by the actual distribution of the data and is less sensitive to outliers.

2.4 RCP CO_2 emission scenarios

The representative concentration pathways (RCP) is a database that documents the emissions, concentrations, and land-cover change projections. The documentation of RCPs can be found at the website https://tntcat.iiasa.ac.at/RcpDb/dsd?Action=htmlpage&page=welcome#intro.

The RCP 2.6 (Van Vuuren et al. 2007) is representative for scenarios in the literature leading to very low greenhouse gas concentration levels. It is a so-called "peak" scenario: its radiative forcing level first reaches a value around 3.1 W/m^2 mid-century, returning to 2.6 W/m^2 by 2100. In order to reach such radiative forcing levels, greenhouse gas emissions (and indirectly emissions of air pollutants) are reduced substantially over time.

The RCP 4.5(Smith and Wigley 2006; Clarke et al. 2007) is a stabilization scenario where total radiative forcing is stabilized before 2100 by employment of a range of technologies and strategies for reducing greenhouse gas emissions.

The RCP 6.0 (Fujino et al. 2006; Hijioka et al. 2008) is a stabilization scenario where total radiative forcing is stabilized after 2100 without overshoot by employment of a range of technologies and strategies for reducing greenhouse gas emissions.

The RCP 8.5 (Riahi et al. 2007) is characterized by increasing greenhouse gas emissions over time representative for scenarios in the literature leading to high greenhouse gas concentration levels. The underlying scenario drivers and resulting development path are based on the A2r of IPCC scenarios.

Bias corrected PRECIS RCM weather data for the period 2014–2090 was arranged in DSSAT format for generating a weather file. Daily solar radiation data for the period 01/01/2014–31/12/2090 uses the Hargreaves and Samani Method. This solar radiation generation was verified in the previous article (Pranuthi and Tripathi 2011).

The X build module which is the experimental module of DSSAT takes the data of crop management practices. This module contains data about field operations (planting, harvesting, irrigation, fertilization, and residue placement) which are specified by the users. The crop management practices adopted by most of the farmers of Haridwar district of Uttarakhand were used for running the model. The data was collected by organizing farmer meetings and interacting with them personally. The planting date was set to 7th July as 80–90% of transplanting in the study area is completed by 1st week of July. Irrigations of 100 mm per irrigation are given at every 15–20 days interval to avoid stress over the crop. As far as fertilizer management is concerned, 100 kgs/ha of nitrogen is given as input and for remaining nutrients the model is not sensitive to them.

3 Results and discussion

3.1 Calibration and validation of PRECIS RCM bias corrected data

The PRECIS original data (RCM) was biased from the observed data which made it unusable for further studies. So, to make it usable, the bias correction methods had to be employed for reliable crop yield forecasts. For calibration, weather data from 1979 to 2009 was selected as training set. The bias corrected data was evaluated using various statistical measures and the evaluation results between observed, RCM predicted and bias corrected RCM data during the calibration period are presented in Table 1. The observed, RCM predicted, and bias corrected RCM average annual rainfall are 985.2, 755.4, and 1021.0 mm respectively during the

calibration period. The observed, RCM predicted, and bias corrected RCM mean maximum temperatures are 30.0, 31.1, and 30.0 °C respectively during the period 1979–2009. The observed, RCM predicted, and bias corrected RCM mean minimum temperatures are 17.4, 19.2, and 17.35 °C respectively during the period 1979–2009. Standard deviation of observed, RCM predicted and bias corrected RCM average annual rainfall are 116.5, 89.0, and 127.3 mm respectively. Standard deviation of observed, RCM predicted and bias corrected RCM average annual rainfall are 5.7, 8.6, and 5.8 °C respectively. Standard deviation of observed, RCM predicted and bias corrected RCM average annual rainfall are 7.0, 8.9, and 7.15 °C respectively. The MBE of RCM predicted rainfall, maximum, and minimum temperatures from the observed values is − 19.2 mm, and 1.0 and 1.8 °C respectively. The MBE of bias corrected rainfall, maximum, and minimum temperatures from the observed values is 2.9 mm, and − 0.04 and − 0.03 °C respectively. The NMSE calculated between observed and RCM predicted rainfall, maximum, and minimum temperatures is 0.007, 0.15 and 0.08 respectively. The NMSE calculated between observed and bias corrected rainfall, maximum, and minimum temperatures is 0.008, 0.1 and 0.05 respectively. The results from t-test and F test showed that the RCM predicted rainfall data is significantly different from the observed data.

During validation period (2010–2013) the observed, RCM predicted, and bias corrected RCM average annual rainfall are 1249.4 mm, 863.7 mm and 1237.5 mm respectively (Table 2). The observed, RCM predicted, and bias corrected RCM mean maximum temperatures are 29.9, 31.7, and 29.8 °C respectively. The observed, RCM predicted, and bias corrected RCM mean minimum temperatures are 18.3, 19.4, and 18.4 °C respectively. Standard deviation of observed, RCM predicted, and bias corrected RCM average annual rainfall are 168.9, 115.5, and 182.9 mm respectively. Standard deviation of observed, RCM

Table 1 Observed, RCM, and bias corrected RCM monthly data of rainfall, Tmax, and Tmin during the period 1/1/1979–31/12/2009

Month	Rainfall (mm)			Tmax (°C)			Tmin (°C)		
	Obs	RCM	RCM Corr	Obs	RCM	RCM Corr	Obs	RCM	RCM Corr
Jan	29.0	24	29.4	20.0	18.6	21.6	6.8	5.1	5.98
Feb	36.9	14	16.3	23.2	24.9	25.9	9.4	9.2	9.27
Mar	25.8	23	27.6	28.6	32.5	30.9	13.5	16.4	15.08
Apr	17.9	16	18.0	34.9	38.9	35.3	18.2	22.7	20.23
May	33.6	12	13.0	37.3	45.3	39.6	22.7	29.9	26.00
Jun	104.1	85	115.6	36.3	42.6	37.8	25.1	31.4	27.19
Jul	273.4	264	373.8	33.6	32.9	31.2	25.7	26.8	23.48
Aug	273.7	194	272.9	32.9	31.7	30.4	25.4	25.9	22.82
Sep	148.7	56	73.7	32.6	32.3	30.8	23.6	24.1	21.36
Oct	23.1	21	26.0	31.3	30.1	29.3	18.1	19.2	17.35
Nov	5.8	26	32.0	27.2	24.2	25.4	12.4	12.8	12.16
Dec	13.2	20	22.5	22.4	18.8	21.8	7.9	6.8	7.34
Total/avg	*985.2*	*755.4*	*1021.0*	*30.0*	*31.1*	*30.0*	*17.4*	*19.2*	*17.35*
SD	*116.5*	*89.0*	*127.3*	*5.7*	*8.6*	*5.8*	*7.0*	*8.9*	*7.15*
MBE		*− 19.2*	*2.9*		*1.0*	*− 0.04*		*1.8*	*− 0.03*
NMSE		*0.007*	*0.008*		*0.15*	*0.10*		*0.08*	*0.05*
Z test		*2.52**	*− 0.33*		*− 1.91*	*0.10*		*− 3.06**	*0.06*
F test		*1.71**	*0.84*		*0.45*	*0.98*		*0.63*	*0.96*

The italicized values are the results of evaluation tests performed to evaluate the bias corrected PRECIS RCM data

*Indicates significance at 0.05 level of significance

predicted, and bias corrected RCM average annual rainfall are 6.0, 8.3, and 6.1 °C respectively. Standard deviation of observed, RCM predicted, and bias corrected RCM average annual rainfall for the period 2010–2013 are 6.6, 8.5, and 6.7 °C respectively.

3.2 Evaluation of bias corrected PRECIS RCM data

The evaluation of bias corrected PRECIS RCM data for the validation period (2010–2013) is presented in Table 2. The MBE of RCM predicted rainfall, maximum, and minimum temperatures from the observed values is − 32.1 mm, and 1.7 and 1.2°C respectively. The MBE of bias corrected rainfall, maximum, and minimum temperatures from the observed values is − 0.9 mm, and − 0.2 °C and 0.1 °C respectively. The NMSE calculated between observed and RCM predicted rainfall, maximum, and minimum temperatures is 0.004, 0.1, and 0.09 respectively. The NMSE calculated between observed and bias corrected rainfall, maximum, and minimum temperatures is 0.005, 0.08, and 0.06 respectively. *t* test results showed that the RCM predicted maximum and minimum temperature data is significantly different from the observed values. *F* test results indicate that the rainfall predicted by PRECIS RCM is significantly different from the observed values. The *t* test and *F* test values showed that the bias corrected RCM data and observed are not significantly different from each other.

The hit score calculated for RCM predicted and bias corrected RCM rainfall data during calibration and validation are given in Table 3. The hit score for RCM predicted rainfall data and bias corrected RCM rainfall data during calibration period is 0.35 and 0.69 respectively. From this score, it is evident that there is improvement in the FA in bias corrected RCM (0.74) than the RCM predicted (0.69) rainfall data during validation period.

Table 2 Observed, RCM, and bias corrected RCM monthly data of rainfall, Tmax, and Tmin during the period 1/1/2010–31/12/2013

Month	Rainfall (mm)			Tmax (°C)			Tmin (°C)		
	Obs	RCM	RCM Corr	Obs	RCM	RCM Corr	Obs	RCM	RCM Corr
Jan	29.7	18.3	21.8	18.2	20.2	21.3	6.7	5.6	7.39
Feb	59.3	28.7	28.5	23.5	24.4	24.4	12.1	10.3	11.11
Mar	7.0	4.5	1.8	30.0	31.8	29.9	16.0	15.7	15.37
Apr	8.0	10.9	6.7	34.6	38.1	34.5	19.5	22.4	20.72
May	37.6	10.5	10.2	37.7	44.6	39.3	23.3	28.8	25.83
Jun	135.0	35.8	43.0	35.9	44.7	39.4	25.5	31.9	28.25
Jul	364.6	291.6	451.7	32.8	34.7	32.0	25.5	26.8	24.23
Aug	410.5	278.9	431.1	32.4	31.9	29.9	24.8	25.9	23.49
Sep	170.3	122.1	178.8	33.1	32.1	30.1	24.1	23.5	21.64
Oct	6.6	6.1	0.1	31.8	31.5	29.7	19.3	19.7	18.62
Nov	0.5	23.9	27.9	27.7	25.2	25.0	13.7	14.1	14.14
Dec	20.6	32.6	36.0	21.9	21.2	22.0	8.7	8.5	9.70
Total/avg	*1249.4*	*863.7*	*1237.5*	*29.9*	*31.7*	*29.8*	*18.3*	*19.4*	*18.4*
SD	*168.9*	*115.5*	*182.9*	*6.0*	*8.3*	*6.1*	*6.6*	*8.5*	*6.7*
MBE		*− 32.1*	*− 0.9*		*1.7*	*− 0.2*		*1.2*	*0.1*
NMSE		*0.004*	*0.005*		*0.1*	*0.08*		*0.09*	*0.06*
t test		*1.78*	*0.05*		*− 2.59**	*0.35*		*−2.21**	*− 0.26*
F test		*2.14**	*0.85*		*0.53*	*0.98*		*0.61*	*0.96*

The italicized values are the results of evaluation tests performed to evaluate the bias corrected PRECIS RCM data

*Indicates significance at 0.05 level of significance

Table 3 Hit score of RCM and bias corrected RCM monthly rainfall for the period 01/01/1979–31/12/2013

Months	Hit score of RCM		Hit score of bias corrected RCM	
	1979–2009	2010–2013	1979–2009	2010–2013
Jan	0.33	0.63	0.67	0.73
Feb	0.50	0.68	0.66	0.67
Mar	0.39	0.81	0.70	0.84
Apr	0.38	0.72	0.77	0.81
May	0.57	0.77	0.71	0.79
Jun	0.41	0.53	0.60	0.6
Jul	0.42	0.44	0.46	0.48
Aug	0.47	0.45	0.51	0.52
Sep	0.25	0.49	0.59	0.67
Oct	0.13	0.89	0.89	0.93
Nov	0.15	0.74	0.91	0.93
Dec	0.23	0.75	0.86	0.9
Annual	*0.35*	*0.69*	*0.69*	*0.74*

The bias correction of mean and standard deviation resulted in decreased mean bias error. For maximum temperatures, the MBE was brought down from 1.0 to − 0.04 and for minimum temperatures, it is brought down from 1.8 to 0.03. Even NMSE was reduced to 0.10 from 0.15 and 0.05 from 0.08 in the case of maximum and minimum temperatures respectively. But in the case of rainfall, the NMSE increased from 0.007 to 0.008 but the MBE decreased from − 19.2 to 2.9. Even during validation period, bias correction decreased the MBE and NMSE of RCM data from 1.2 to 0.1 and 0.09 to 0.06 respectively for minimum temperature. MBE and NMSE for temperatures indicated that the bias correction method applied was good, which can also be applied for the remaining of the data. The applicability of bias correction method to rainfall data cannot be ascertained due to inconsistency in the values of MBE and NMSE.

The monthly hit score shows that bias corrected data FA is better during all the months during calibration period. Even during validation, the bias corrected PRECIS forecast showed higher hit rate than the original PRECIS data except for January, February, March, and May months but overall FA is higher for bias corrected PRECIS rainfall data. The bias correction method used in this study is a combination of two methods. The method suggested by (Hay et al. 2002) was used to bias correct rainfall and method suggested by (Leander and Buishand 2007) was used to bias correct maximum and minimum temperatures. The first method is used to correct the mean of precipitation and second method is used to correct coefficient of variation (CV) of maximum and minimum temperatures. Both these methods are combined to bias correct the mean and CV of rainfall, maximum, and minimum temperatures in this study. However, the method suggested in this research can be applied to bias correct any climatic parameter as such. The reduction in bias of PRECIS RCM data was reportedly decreased after employing various bias correction methods (Akhtar et al. 2009; Teutschbein and Seibert 2012).

3.3 Mann-Kendall's trend analysis

3.3.1 Annual trend

Trend analysis of future rainfall, maximum, and minimum temperatures predicted by PRECIS RCM (Table 4) revealed that increase in annual rainfall and temperatures is significant. Theil Sen's slope

Table 4 Mann-Kendall's test and Thiel Sen's slope for rainfall, Tmax, and Tmin for the monthly forecasted data (2014–2090)

Month	Rainfall (2014–2090)			Tmax (2014–2090)			Tmin (2014–2090)		
	Z	Sig	Sen's slope	*Z*	Sig	Sen's slope	*Z*	Sig	Sen's slope
Jan	−1.05		− 0.041	*3.58*	***	*0.051*	*4.74*	***	*0.032*
Feb	0.79		0.012	*3.03*	**	*0.041*	*5.57*	***	*0.043*
Mar	0.69		0.019	*2.23*	*	*0.025*	*4.41*	***	*0.030*
Apr	*− 1.99*	*	*− 0.065*	*5.01*	***	*0.050*	*4.25*	***	*0.030*
May	0.05		0.000	*5.31*	***	*0.045*	*4.74*	***	*0.027*
Jun	0.78		0.370	*2.97*	**	*0.039*	*4.26*	***	*0.028*
Jul	*4.40*	***	*3.310*	*5.98*	***	*0.041*	*9.00*	***	*0.042*
Aug	*4.68*	***	*3.161*	*7.20*	***	*0.045*	*9.43*	***	*0.038*
Sep	−1.23		− 0.632	*7.02*	***	*0.048*	*8.40*	***	*0.042*
Oct	*− 1.81*	+	*− 0.001*	*6.60*	***	*0.058*	*6.88*	***	*0.061*
Nov	− 0.68		0.000	*4.63*	***	*0.039*	*5.39*	***	*0.044*
Dec	0.62		0.004	*4.15*	***	*0.050*	*5.99*	***	*0.040*
Kharif	*3.91*	***	*6.005*	*7.32*	***	*0.048*	*8.76*	***	*0.043*
Annual	*3.97*	***	*5.828*	*7.47*	***	*0.046*	*9.50*	***	*0.039*

Italicized values indicate significance of trend at any level of significance

*Significant trend at 0.05 level of significance, +Significant trend at 0.1 level of significance

**Significant trend at 0.01 level of significance

***Significant trend at 0.001 level of significance

estimated for PRECIS weather by 2090 is 5.828 mm/year, and 0.046 and 0.039 °C/year for annual rainfall, maximum, and minimum temperatures respectively (Fig. 1). After the completion of correction and trend analysis of PRECIS RCM data, it can be concluded that the rainfall would increase up to 450 mm and maximum and minimum temperatures by 3.6 and 3.0 °C by the end of 2090.

3.3.2 Seasonal trend

The increase in rainfall (462 mm) during rainy months (*Kharif*) is more prominent and is likely to impact the yields of rice crop. Even the alteration in temperatures would affect the development of rice crop as it is expected that the temperature during the crop growth period (June–October) is likely to increase by 3.7 and 3.3 °C for maximum and minimum temperatures respectively (Fig. 2). The change in the rainfall amounts and temperature would also alter many other weather parameters such as humidity and evaporation wind speed. which would in turn directly or indirectly affect the crop growth and development.

For Haridwar district, the PRECIS RCM predicted that annual rainfall, maximum, and minimum temperatures are likely to increase by 2090. Increasing GHG concentrations are the most probable reason that could be attributed to this increase. PRECIS RCM predicted that all-India annual mean temperatures would increase by 3.5–4.3 °C by the end of this century. Similarly, all-India annual precipitation would increase by 9–16% with negative trend towards south and western states according to a study done by (Kumar et al. 2006; Akhtar et al. 2009).

3.4 DSSAT CERES rice simulation under various RCP scenarios

Rice productivity was predicted for 2014–2090 presuming no constraint of insect pest and disease, soil fertility, water, and management constraint under various RCP predicted CO_2

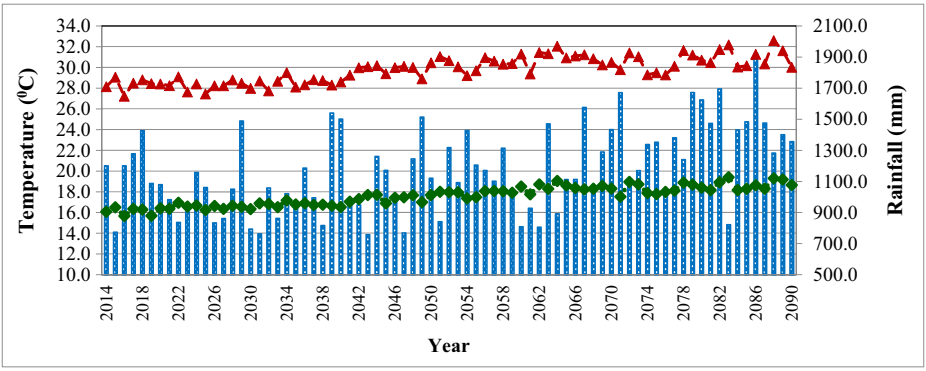


Fig. 1 Forecasted annual rainfall and annual average Tmax, Tmin, and SRAD for Haridwar district during 2014–2090

emission scenarios, and its average per decade is presented in Table 5. The simulation result showed that if the crop is grown with the existing soil and crop management conditions, rice productivity of Haridwar in general will decrease by 31.7 kgs/ha/year. The average grain yield of rice during 2020–2090 under present CO_2 scenario is 5586 kgs/ha. Under the present scenario, assuming no rise in atmospheric CO_2 concentrations, the rice productivity will decline with increasing temperature.

The yield simulated by CERES rice indicated that the rice yield of Haridwar district would decrease by 2099 kgs/ha by the end of 2090. To estimate the role of climate in recent yield trends, we applied the regression models to observe the effect of weather parameters on grain yield. The yield weather regression equations at current CO_2 levels showed that the yield majorly depended on the maximum temperature. The decrease in rice yields with no CO_2 elevation (present emission scenario) could be attributed to the climate change and decreased rice yields from increased maximum temperature which has also been reported by (Seshu and Cady 1984). Grain yield declined by 459.7 kgs/ha for each 1 °C increase in growing-season maximum temperature, whereas the effect of minimum temperature on crop yield was insignificant (Peng et al. 2004).

In the case of RCP 2.6 scenario, the increase in atmospheric CO_2 concentration is 0.22 ppm/year; the yield decreased at 30 kgs/ha/year due to climate change. The step-down regression equations indicate that CO_2 concentrations had a positive effect on the rice grain

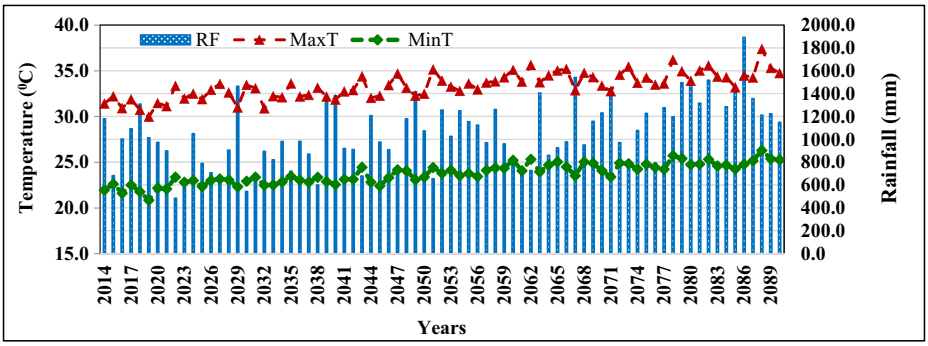


Fig. 2 Forecasted *Kharif* rainfall and *Kharif* average Tmax, Tmin, and SRAD for Haridwar district during 2014–2090

Table 5 Decadal average of yield and weather for that corresponding decade simulated by CERES rice model under RCP emission scenarios for the period 2020–2090 for rice crop cv. Sharbati

Decade	Yield (kgs/ha)					Weather parameters				
	CO_2 emission scenarios					Tmax	Tmin	SRAD	RF	ET
	Current scenario	RCP 2.6	RCP 4.5	RCP 6.0	RCP 8.5	(°C)	(°C)	(MJ/m^2/day)	(mm)	(mm)
2020	7685.0	7817.6	7815.0	7807.6	7830.9	31.2	23.6	18.3	882.2	427.3
2030	7443.8	7698.1	7704.4	7684.0	7764.2	33.1	24.4	20.0	496.4	444.5
2040	7275.2	7613.2	7687.6	7636.6	7816.5	32.2	24.2	19.1	924.0	431.5
2050	6881.1	7273.3	7495.3	7429.3	7758.8	32.6	25.0	18.5	601.1	439.4
2060	6602.3	6993.1	7379.4	7348.3	7805.8	34.7	26.5	19.5	602.9	466.8
2070	6169.0	6544.3	7086.7	7182.1	7813.3	33.4	26	18.5	826.8	417.8
2080	5910.5	6247.7	6892.3	7179.2	7921.5	34.4	26.8	19.1	1104.3	408.9
2090	5586.1	5883.7	6583.2	7089.1	7928.3	34.8	27.2	19.0	908.3	426.2
Mean	6694.1	7817.6	7815.0	7807.6	7830.9	33.3	25.5	19.0	793.3	432.8

yield while the negative effect of PRECIS RCM predicted maximum temperature is more evident than other weather parameters. In these equations, time is assumed as the technological advancements such as use of better varieties and inputs. But even it proved to have a negative impact on yields of rice crop, similar results have been obtained for RCP 4.5 scenario with positive impact of atmospheric CO_2 concentrations and maximum temperature and technology trend (time in years) have a negative impact. In the case of RCP 4.5 emission scenario, the atmospheric CO_2 increases at approximately 2 ppm/year but the yield decreases with a negative slope of 18.2 kgs/ha/year.

According to RCP 6.0 and 8.5 emission scenarios, the atmospheric CO_2 concentrations increase at the rate of 3 and 6 ppm/year respectively. The effect of PRECIS RCM predicted maximum temperature is more pronounced, and the effect of atmospheric CO_2 and other weather parameters is negligible Table 6. It is observed that at higher CO_2 concentrations the yield response to CO_2 is little.

The role of temperature and CO_2 on rice crop yields has been indicated through regression equations (Table 6), so a graph (Fig. 3) showing relation between the cumulative grain yield (the grain yield simulated under various RCP scenarios are pooled) simulated under varying CO_2 concentrations and temperatures is drawn to understand completely their effect on yields. It is observed that the CO_2 has a positive effect on yields and increases the tolerance limits of the crop towards climate change especially increase in atmospheric temperatures (global warming) and leads to sustained yields. This CO_2 fertilization effect has also simulated for

Table 6 Step-down regression equations between CERES rice simulated yields and corresponding weather parameters and CO_2 concentrations for the period 2014–2090 for rice crop cv. Sharbati

Scenario	Equation	R^2
Current CO_2 concentrations	$Y = 16{,}933.6 - 459.7 \times \text{Tmin}$	0.84
RCP 2.6	$Y = 46{,}359.3 - 16.6 \times \text{Time} - 281.9 \times \text{Tmax} + 9.15 \times CO_2$ conc	0.88
RCP 4.5	$Y = 60{,}305.8 - 23.8 \times \text{Time} - 280.8 \times \text{Tmax} + 10.7 \times CO_2$ conc	0.76
RCP 6.0	$Y = 14{,}860.5 - 225.8 \times \text{Tmax}$	0.58
RCP 8.5	$Y = 9836 - 60.8 \times \text{Tmax}$	0.53

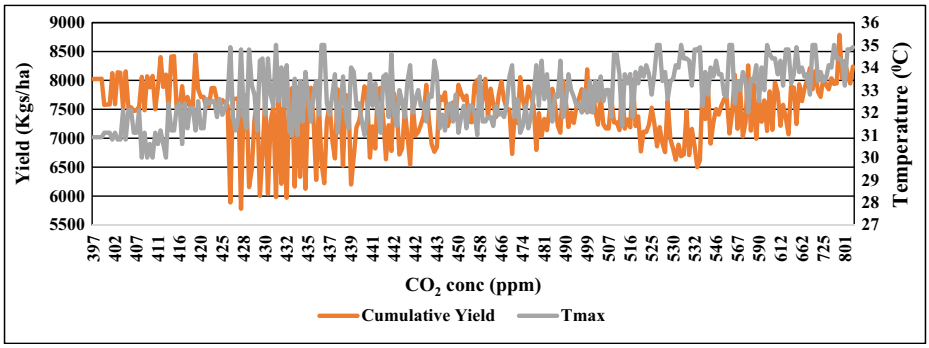


Fig. 3 Graph showing the relation between rice crop yield of Sharbati cv., maximum temperature, and CO_2 concentrations

rice, globally by Christopher Muller (Müller et al. 2015) in which 81% of the negative effect of climate change is by CO_2 fertilization effect. The yield increased with increasing CO_2 concentrations up to 32 °C maximum temperatures. High temperatures and low CO_2 concentrations lead to decreased yield while the yields are sustained under higher temperatures and high CO_2 concentrations. But it is observed from the simulated data that response of the rice crop yield of Sharbati cultivar is less beyond 450 ppm and it further decreases with increasing CO_2 concentrations which is evident from the step-down regression equations given in Table 6. This could be attributed to the acclimation process that occurs when crop is exposed for longer duration of higher CO_2 concentrations which is also reported in rice field experiments (FACE experiments) (Zhu et al. 2012; Tausz et al. 2013; Adachi et al. 2014).

4 Conclusion

The statistical bias correction method adopted to correct PRECIS RCM data was found suitable for bias correction with NMSE less than 0.1. Trend analysis of RCM data for the period 2014–2090 showed that the rainfall would increase at 6.005 mm/year, maximum temperature at 0.048 °C/year, and minimum temperature at 0.043 °C/year for *Kharif* season. DSSAT CERES rice simulations under various RCP CO_2 emission scenarios give an indication that up to certain extent of atmospheric CO_2 concentrations (approximately 450 ppm) the rice yields are positively impacted while the rice yields at higher CO_2 concentrations are not affected much, but the tolerance limit of the crop towards higher temperature are raised. This concludes that the higher CO_2 concentrations in the future will have a positive impact on rice crop especially in the case of grain yield of Sharbati cv. of Haridwar district of Uttarakhand.

References

Adachi M, Hasegawa T, Fukayama H et al (2014) Soil and water warming accelerates phenology and down-regulation of leaf photosynthesis of rice plants grown under free-air CO_2 enrichment (FACE). Plant Cell Physiol 55:370–380

Ahsan S, Ali MS, Hoque MR et al (2010) Agricultural and environmental changes in Bangladesh in response to global warming. In: Climate change and food security in South Asia. Springer, pp 119–134

Akhtar M, Ahmad N, Booij MJ (2009) Use of regional climate model simulations as input for hydrological models for the Hindukush-Karakorum-Himalaya region. Hydrol Earth Syst Sci 13:1075
Clarke L, Edmonds J, Jacoby H et al (2007) Scenarios of greenhouse gas emissions and atmospheric concentrations. US Dep Energy Publ 6
Field CB, Barros VR, Mach K, Mastrandrea M (2014) Climate change 2014: impacts, adaptation, and vulnerability. Cambridge University Press Cambridge and New York
Fowler HJ, Blenkinsop S, Tebaldi C (2007) Linking climate change modelling to impacts studies: recent advances in downscaling techniques for hydrological modelling. Int J Climatol 27:1547–1578
Fujino J, Nair R, Kainuma M et al (2006) Multi-gas mitigation analysis on stabilization scenarios using AIM global model. Energy J:343–353
Gornall J, Betts R, Burke E et al (2010) Implications of climate change for agricultural productivity in the early twenty-first century. Philos Trans R Soc B Biol Sci 365:2973–2989
Grotch SL, MacCracken MC (1991) The use of general circulation models to predict regional climatic change. J Clim 4:286–303
Harley CDG, Randall Hughes A, Hultgren KM et al (2006) The impacts of climate change in coastal marine systems. Ecol Lett 9:228–241
Hay LE, Clark MP, Wilby RL et al (2002) Use of regional climate model output for hydrologic simulations. J Hydrometeorol 3:571–590
Hijioka Y, Matsuoka Y, Nishimoto H et al (2008) Global GHG emission scenarios under GHG concentration stabilization targets. J Glob Environ Eng 13:97–108
Jones JW, Hoogenboom G, Porter CH et al (2003) The DSSAT cropping system model. Eur J Agron 18:235–265
Kendall MG (1946) The advanced theory of statistics. Adv theory Stat
Kumar KR, Sahai AK, Kumar KK et al (2006) High-resolution climate change scenarios for India for the 21st century. Curr Sci:334–345
Leander R, Buishand TA (2007) Resampling of regional climate model output for the simulation of extreme river flows. J Hydrol 332:487–496
Lenderink G, van Ulden A, van den Hurk B, Keller F (2007) A study on combining global and regional climate model results for generating climate scenarios of temperature and precipitation for the Netherlands. Clim Dyn 29:157–176
Mann HB (1945) Nonparametric tests against trend. Econom J Econom Soc:245–259
Maraun D, Wetterhall F, Ireson AM et al (2010) Precipitation downscaling under climate change: recent developments to bridge the gap between dynamical models and the end user. Rev Geophys 48
Müller C, Elliott J, Chryssanthacopoulos J et al (2015) Implications of climate mitigation for future agricultural production. Environ Res Lett 10:125004
Pranuthi G, Tripathi SK (2011) Development of Hargreaves and Samani shortwave radiation coefficient (Kr) for India. J Agrometeorol 13:150–152
Rao AVMS, Shanker AK, Rao VUM et al (2016) Predicting irrigated and rainfed rice yield under projected climate change scenarios in the eastern region of India. Environ Model Assess 21:17–30
Riahi K, Grübler A, Nakicenovic N (2007) Scenarios of long-term socio-economic and environmental development under climate stabilization. Technol Forecast Soc Change 74:887–935
Roudier P, Sultan B, Quirion P, Berg A (2011) The impact of future climate change on West African crop yields: what does the recent literature say? Glob Environ Chang 21:1073–1083
Satapathy SS, Swain DK, Herath S (2014) Field experiments and simulation to evaluate rice cultivar adaptation to elevated carbon dioxide and temperature in sub-tropical India. Eur J Agron 54:21–33
Sen PK (1968) Estimates of the regression coefficient based on Kendall's tau. J Am Stat Assoc 63:1379–1389
Singh PK, Singh KK, Bhan SC et al (2017) Impact of projected climate change on rice (Oryza sativa L.) yield using CERES-rice model in different agroclimatic zones of India. Curr Sci 112:108
Smith SJ, Wigley TML (2006) Multi-gas forcing stabilization with Minicam. Energy J:373–391
Solomon S, Qin D, Manning M, et al (2007) Climate change 2007: the physical science basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change
Srivastava P, Singh R, Tripathi S, Raghubanshi AS (2016) An urgent need for sustainable thinking in agriculture—an Indian scenario. Ecol Indic 67:611–622
Subash N, Singh SS, Priya N (2013) Observed variability and trends in extreme temperature indices and rice–wheat productivity over two districts of Bihar, India—a case study. Theor Appl Climatol 111:235–250
Tausz M, Tausz-Posch S, Norton RM et al (2013) Understanding crop physiology to select breeding targets and improve crop management under increasing atmospheric CO_2 concentrations. Environ Exp Bot 88:71–80
Teutschbein C, Seibert J (2012) Bias correction of regional climate model simulations for hydrological climate-change impact studies: review and evaluation of different methods. J Hydrol 456:12–29

Theil H (1992) A rank-invariant method of linear and polynomial regression analysis. In: Henri Theil's contributions to economics and econometrics. Springer, pp 345–381

Tripathi SK, Singh M, Pandey A (2007) Agroclimatic variability analysis of Roorkee, Uttarakhand. J IWRS 27(3 4):24–30

Van Vuuren DP, Den Elzen MGJ, Lucas PL et al (2007) Stabilizing greenhouse gas concentrations at low levels: an assessment of reduction strategies and costs. Clim Chang 81:119–159

Wood AW, Maurer EP, Kumar A, Lettenmaier DP (2002) Long-range experimental hydrologic forecasting for the eastern United States. J Geophys Res Atmos 107

Yu W, Alam M, Hassan A et al (2010) Climate change risks and food security in Bangladesh. Routledge

Zhu C, Ziska L, Zhu J et al (2012) The temporal and species dynamics of photosynthetic acclimation in flag leaves of rice (Oryza sativa) and wheat (Triticum aestivum) under elevated carbon dioxide. Physiol Plant 145:395–405

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